# Jetson Orin Nano Developer Kit Carrier Board

Specification

SP-11324-001_v1.3 | December 2024

# Document History

SP-11324-001_v1.3

| Version | Date | Description of Change |
|---|---|---|
| 1.0 | March 20, 2023 | Initial release |
| 1.1 | May 17, 2023 | > Updated Note in Chapter 1: Introduction with VDD_IN related information.<br>> Updated Pin #13 in Table 2-5: DisplayPort Connector Pin Description – J8.<br>> Updated Pin #55 and Pin #57 in Table 2-6: M.2 Key E Expansion Slot Pin Description – J10.<br>> Added developer kit weight.<br>> Updated Figure 5-1: Power Diagram.<br>> Updated VDD_5V_SYS in Table 5-2: Interface Supply Current Capabilities.<br>> Updated Table 5-3: Supply Current Capabilities per Connector per Supply; replaced VDD_IN with VDD_5V_SYS. |
| 1.2 | April 12, 2024 | > Updated Figure 1-4. Jetson Orin Nano Carrier Board Placement – Top View: corrected Pin 1 location on J20 and J21 camera connectors |
| 1.3 | December 20. 2024 | > Introduction: updated note - Orin NX 40W (MAXN_SUPER) is NOT supported on the developer kit carrier board.<br>> Jetson Orin Nano Module Feature List: updated Memory to reflect MAXN_SUPER performance<br>> CAN Bus Header: corrected typo; reference to a 2.45 mm pitch header (J17) was corrected to say 2.54 mm pitch header (J17) |

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# Chapter 1. Introduction

This specification contains recommendations and guidelines for engineers to follow to create modules for the expansion connectors on the NVIDIA® Jetson Orin™ Nano carrier board. As well as understand the capabilities of the other dedicated interface connectors and associated power solutions on the platform.

The Jetson Orin Nano carrier board is ideal for software development within the Linux environment. Standard connectors are used to access Jetson Orin Nano features and interfaces, enabling a highly flexible and extensible development platform. Go to https://developer.nvidia.com/embedded/develop or r contact your NVIDIA representative for access to software updates and the developer SDK supporting the OS image and host development platform that you want to use. The developer SDK includes an OS image that you will load onto your Jetson Orin Nano device, supporting documentation, and code samples to help you get started.

The Jetson Orin Nano carrier board supports all the Jetson Orin Nano Series and Jetson Orin NX Series modules.

**Caution:** ALWAYS CONNECT JETSON ORIN NANO and ALL EXTERNAL PERIPHERAL DEVICES BEFORE CONNECTING THE POWER SUPPLY TO THE AC POWER JACK. Connecting a device while powered on may damage the developer kit carrier board, Jetson Orin Nano, or peripheral device. In addition, the carrier board should be powered down and the power removed before plugging or unplugging devices or add-on modules into the headers. Wait for the red power VDD_IN LED to turn off or wait for 5 minutes if your system does not have a power LED. This includes the Jetson Orin Nano module, the camera connector, the M.2 connector, and the other expansion headers.

The Jetson Orin Nano developer board contains ESD-sensitive parts. Always use appropriate anti-static and grounding techniques when working with the system. Failure to do so can result in ESD discharge to sensitive pins, and irreparably damage your Jetson Orin Nano board. NVIDIA will not replace units that have been damaged due to ESD discharge.

**Note:** The developer kit carrier board has been modified to only support 5V to the module. The MODULE_ID signal is not pulled up on the carrier board. This means that regardless of the capability of the module, VDD_IN will be 5V only. Custom carrier boards can still be designed to support 5V and 19V and use the MODULE_ID to identify a 5V vs. 19V Input. Orin NX 40W (MAXN_SUPER) is NOT supported on the developer kit carrier board.

## 1.1 Jetson Orin Nano Module Feature List

The following is a list of features for the Jetson Orin Nano module.

- Applications processor (AP)
  - NVIDIA Orin™
- Memory
  - 8 GB 128-bit wide LPDDR5 DRAM (up to 68 GB/s; 102GB/s MAXN_SUPER)
  - Micro SD card socket (UHS-1)
- Networking
  - 10/100/1000 BASE-T Ethernet
- Advanced power management (APM)
- Dynamic voltage and frequency scaling
- Multiple clock and power domains

## 1.2 Carrier Board Feature List

The following is a list of features for the carrier board.

- Connection to Jetson Orin Nano
  - 260-pin SO-DIMM connector
- USB
  - USBC: Supports Recovery Mode
  - USB 3.2 (Gen2x1) Hub to 4x Type A (host only)
- Wired network
  - Gigabit Ethernet (RJ45 connector with LEDs and optional PoE header)
- Display
  - VESA® DisplayPort™ (DP v1.2 (+MST) and eDP v1.4)
- Camera connectors
  - 2x 22-position flex connectors
  - CSI (2.5 Gbps per pair): 1, x2 or x4 and 1, x2
  - Camera CLK, I2C, and control
- M.2 Key E connector
  - PCIe (Gen3) x1 Lane, USB 2.0
  - I2S, UART, I2C
  - Control

- M.2 Key M connector (x2)
  - #1) PCIe (Gen3) x4 lane, control
  - #2) PCIe (Gen3) x2 lane, control
- Expansion header
  - 40-pin (2x20) header
  - I2C (x2), SPI (x2), UART
  - I2S, audio clock, GPIOs, PWMs
- UI and indicators
  - Button header: Power, reset, force recovery, debug UART, Auto-Power-On disable
  - LEDs: Power
- Miscellaneous
  - Fan connector:  5V, PWM, and tach
  - Optional RTC back-up connector
  - Optional CAN header
- Power
  - DC Jack: 9-20 V (19V supply provided)
  - Optional Ethernet PoE and back power headers
  - Main 5.0 V Pre-regulator: GS9230 (or A0Z2264)
    - Provides VDD_IN to module
  - Main 3.3V supply: GS9230 (or A0Z2264)
  - Main 1.8V supply:  GS7116S5
  - 3.3V AO (always on) supply: GS7116S5
  - USB VBUS load switches:  AP22811AW5-7 (x2)
  - DP 3.3V power switch: APL3552ABI-TRG
- Developer kit operating temperature range
  - 0ºC to 35ºC

## 1.3 Jetson Orin Nano Carrier Board Block Diagram

Figure 1-1 through Figure 1-5 show the block diagram and various placement views for Jetson Orin Nano and the carrier board.

Figure 1-1. Jetson Orin Nano Block Diagram

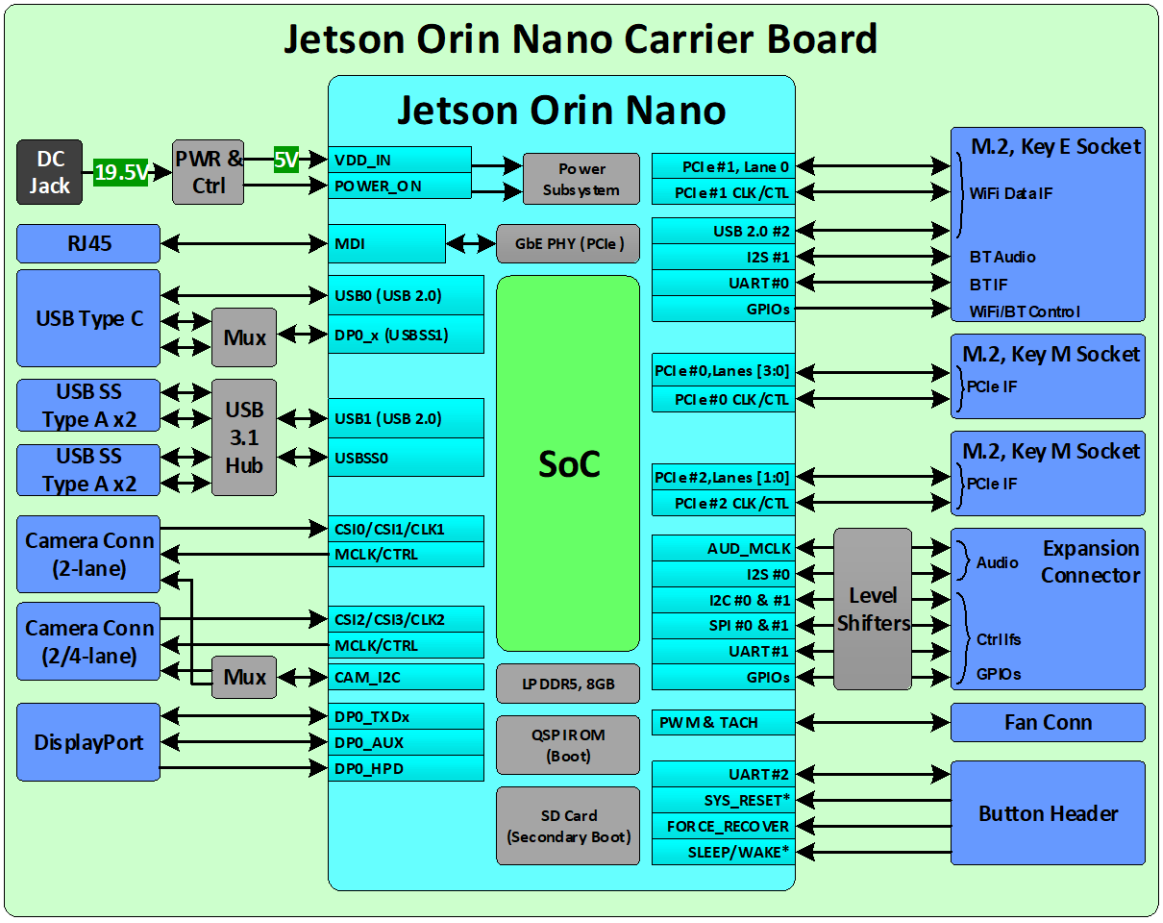
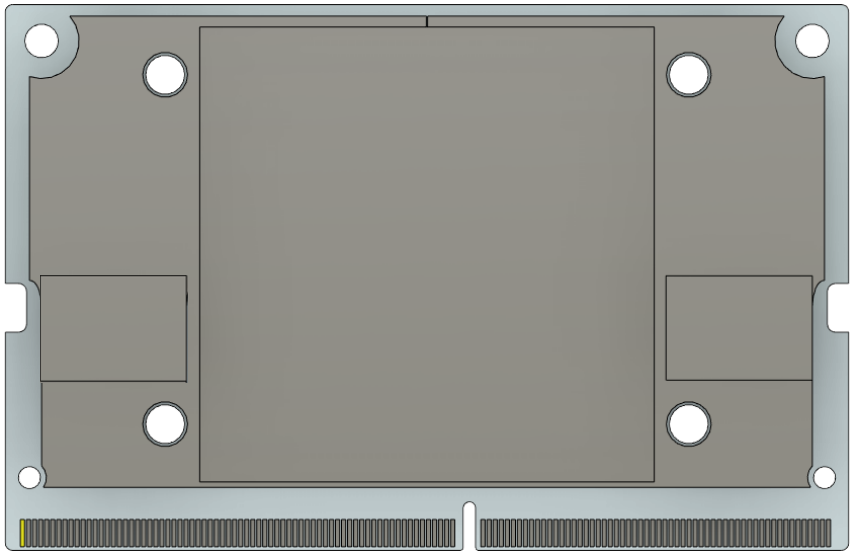
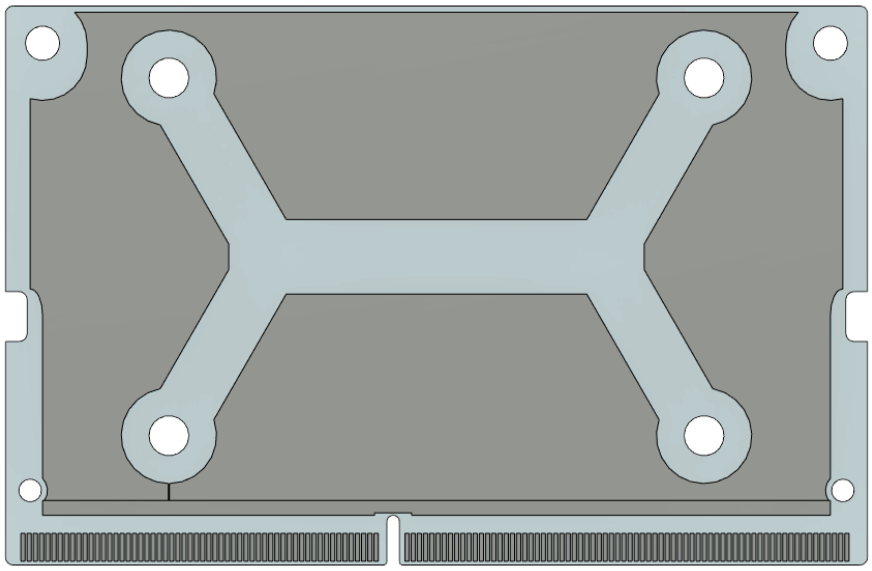


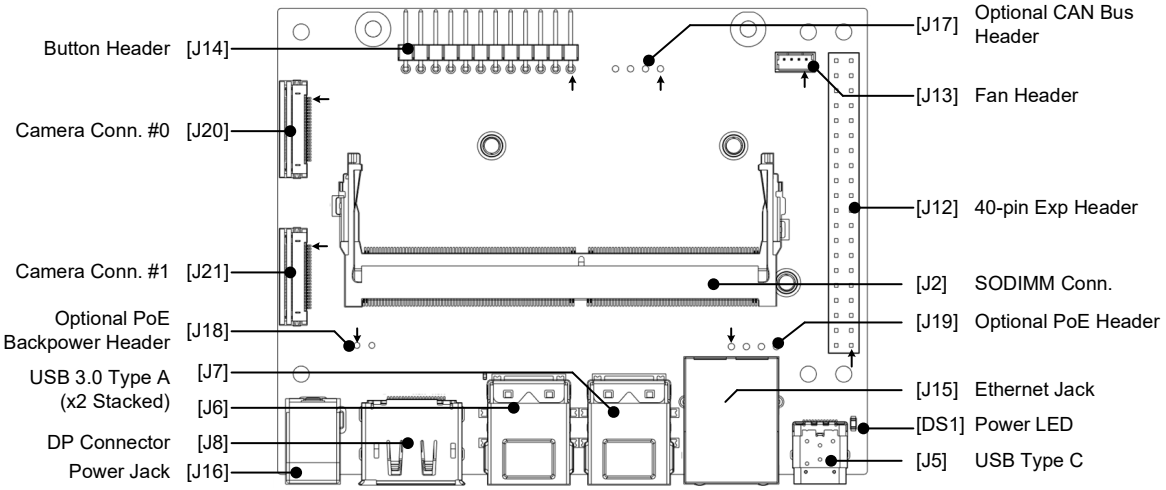

Figure 1-2. Jetson Orin Nano – Top View (Envelope)



## Figure 1-3. Jetson Orin Nano – Bottom View (Envelope)

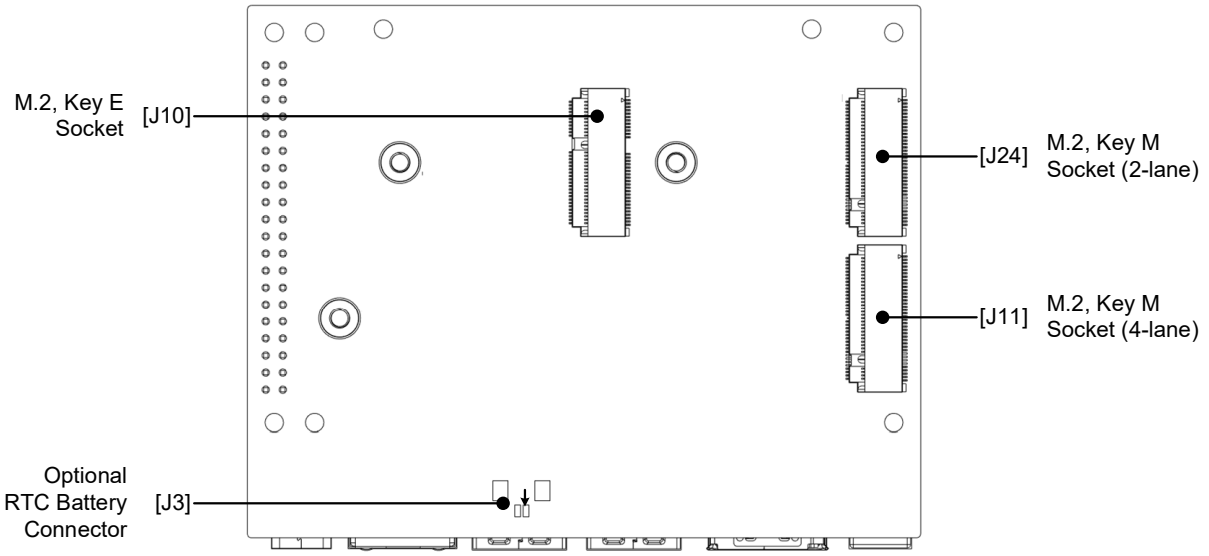


## Figure 1-4. Jetson Orin Nano Carrier Board Placement – Top View




| | | | |
|---|---|---|---|
| **J2** | Jetson Orin Nano Conn. (SODIMM, 260-pin) | **J15** | RJ45 Ethernet Socket, 18-pins, RA, Female |
| **J5** | USB Type C | **J16** | Power Jack |
| **J6** | USB Type A Dual Stacked Connector | **J17** | Optional: CAN Bus Header (1x4, 2.54 mm pitch, RA) |
| **J7** | USB Type A Dual Stacked Connector | **J18** | Optional: PoE Backpower Header (1x2, 2.54 mm pitch) |
| **J8** | DisplayPort Connector | **J19** | Optional: PoE Header (1x4, 2.54 mm pitch) |
| **J12** | 40-pin Expansion Header (2x20, 2.54 mm pitch) | **J20** | Camera (#0) Connector (22 pos, 0.5 mm pitch) |
| **J13** | Fan Header (4-pin, 1.25 mm pitch) | **J21** | Camera (#1) Connector (22 pos, 0.5 mm pitch) |
| **J14** | Button Header (1x12, 2.54 mm pitch, RA) | **DS1** | Power LED (Green) |

## Figure 1-5. Jetson Orin Nano Carrier Board Placement – Bottom View




| | | | |
|---|---|---|---|
| **J3** | Optional: RTC Back-up Coin Cell Socket (CR1225) | **J11** | M.2 Key M Slot (75-pin) |
| **J10** | M.2 Key E Connectivity Slot (75-pin) | **J24** | M.2 Key M Slot (75-pin) |

# Chapter 2. Jetson Nano Carrier Board Standard Connectors

The Jetson Orin Nano carrier board provides several connectors with industry standard pinouts to support additional functionality beyond what is integrated on the main platform board. This includes:

- USB 3.2:  Type C connector
- USB 3.2:  2 x Type A stacked connectors
- Gigabit Ethernet:  RJ45 connector
- DisplayPort connector
- M.2, Key E socket
- M.2, Key M socket (4-lane PCIe)
- M.2, Key M socket (2-lane PCIe)

## 2.1 USB Ports

The carrier board supports five USB connectors. One is a USB 3.2 Type C connector (J5) supporting host, device, and USB Recovery. In addition, there are two, dual stacked USB 3.2 Type A connectors (J6 and J7). Each connector supports host mode only. A load switch supplying VBUS is provided for each of the USB 3.2 ports per stack and is limited to 3A of output current.

**Table 2-1. USB 3.2 Type C Connector Pin Description – J5**

| Pin # | Connector Pin Name | Associated Module Pin Name (See Note 1 and 2) | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|---|
| A1 | GND_A | – | – | Ground | Ground |
| A2 | TX1_P | USBSS1_TX_P | G23 | USB 3.2 #1 Transmit 1 from mux | Output |
| A3 | TX1_N | USBSS1_TX_N | G22 | | |
| A4 | – | – | – | USB VBUS_A Power | Power |
| A5 | CC1 | – | – | CC 1 from CC Controller | Output |
| A6 | D1_P | USB0_D_P | F12 | USB 2.0 #0 Data 1 | Bidir |

| Pin # | Connector Pin Name | Associated Module Pin Name (See Note 1 and 2) | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|---|
| A7 | D1_N | USB0_D_N | F13 | | |
| A8 | SBU1 | – | – | Unconnected | – |
| A9 | – | – | – | USB VBUS_A Power | Power |
| A10 | RX2_N | USBSS1_RX_P | C22 | USB 3.2 #1 Receive 2 from mux | Input |
| A11 | RX2_P | USBSS1_RX_N | C23 | | |
| A12 | GND_A | – | – | Ground | Ground |
| B1 | GND_B | – | – | Ground | Ground |
| B2 | TX2_P | USBSS1_TX_P | G23 | USB 3.2 #1 Transmit 2 from mux | Output |
| B3 | TX2_N | USBSS1_TX_N | G22 | | |
| B4 | – | – | – | USB VBUS_A Power | Power |
| B5 | CC2 | – | – | CC 2 from CC Controller | Output |
| B6 | D2_P | USB0_D_P | F12 | USB 2.0 #0 Data 2 | Bidir |
| B7 | D2_N | USB0_D_N | F13 | | |
| B8 | SBU2 | – | – | Unconnected | – |
| B9 | – | – | – | USB VBUS_A Power | Power |
| B10 | RX1_N | USBSS1_RX_P | C22 | USB 3.2 #1 Receive 1 from mux | Input |
| B11 | RX1_P | USBSS1_RX_N | C23 | | |
| B12 | GND_B | – | – | Ground | Ground |

**Notes:**

1. **USB 3.2 module pin names are using the Orin NX and Orin Nano function names.**
2. **The module pins for the USB 3.2 ports are not directly connected to the USB connector pins but are routed through a multiplexer.**
3. **In the Type/Dir column, Output is to USB connectors. Input is from USB connectors. Bidir is for bidirectional signals.**

| **Legend** | Ground | Power | Reserved |
|---|---|---|---|

## Table 2-2. USB 3.2 Type A Connector Pin Descriptions – J6

| Pin # | Module Pin Name (see Note 1 and 2) | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| **USB 3.0 Type A (2)** | | | | |
| 1 | – | – | VBUS Supply | Power |
| 2 | USB1_D_N | | USB 2.0 #2 Data from hub | Bidir |
| 3 | USB1_D_P | | | |
| 4 | – | – | Ground | Ground |
| 5 | USBSS0_RX_N | 161 | USB 3.1 Receive #2 Data from hub | Input |

| Pin # | Module Pin Name (see Note 1 and 2) | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| 6 | USBSS0_RX_P | 163 | | |
| 7 | – | – | Ground | Ground |
| 8 | USBSS0_TX_N | 166 | USB 3.1 Transmit #2 Data from hub | Output |
| 9 | USBSS0_TX_P | 168 | | |
| **USB 3.0 Type A (1)** | | | | |
| 10 | – | – | VBUS Supply | Power |
| 11 | USB1_D_N | 115 | USB 2.0 Data #1 Data from hub | Bidir |
| 12 | USB1_D_P | 117 | | |
| 13 | – | – | Ground | Ground |
| 14 | USBSS0_RX_N | 161 | USB 3.1 Receive #1 Data from hub | Input |
| 15 | USBSS0_RX_P | 163 | | |
| 16 | – | – | Ground | Ground |
| 17 | USBSS0_TX_N | 166 | USB 3.1 Transmit #1 Data from hub | Output |
| 18 | USBSS0_TX_P | 168 | | |

Notes:

1. USB 3.2 module pin names are using the Orin NX and Orin Nano function names.
2. The module pin names not directly connected to the USB connector pins but are routed to the input of the USB hub.
3. In the Type/Dir column, Output is to USB connectors. Input is from USB connectors. Bidir is for bidirectional signals.

| Legend | Ground | Power | Reserved |
|---|---|---|---|

## Table 2-3. USB 3.2 Type A Connector Pin Description – J7

| Pin # | Module Pin Name (see Note 1 and 2) | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| **USB 3.0 Type A (2)** | | | | |
| 1 | – | – | VBUS Supply | Power |
| 2 | USB1_D_N | 115 | USB 2.0 #4 Data from hub | Bidir |
| 3 | USB1_D_P | 117 | | |
| 4 | – | – | Ground | Ground |
| 5 | USBSS0_RX_N | 161 | USB 3.1 Receive #4 Data from hub | Input |
| 6 | USBSS0_RX_P | 163 | | |
| 7 | – | – | Ground | Ground |
| 8 | USBSS0_TX_N | 166 | USB 3.1 Transmit #4 Data from hub | Output |
| 9 | USBSS0_TX_P | 168 | | |

| Pin # | Module Pin Name (see Note 1 and 2) | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| **USB 3.0 Type A (1)** | | | | |
| 10 | – | – | VBUS Supply | Power |
| 11 | USB1_D_N | 115 | USB 2.0 Data #3 Data from hub | Bidir |
| 12 | USB1_D_P | 117 | | |
| 13 | – | – | Ground | Ground |
| 14 | USBSS0_RX_N | 161 | USB 3.1 Receive #3 Data from hub | Input |
| 15 | USBSS0_RX_P | 163 | | |
| 16 | – | – | Ground | Ground |
| 17 | USBSS0_TX_N | 166 | USB 3.1 Transmit #3 Data from hub | Output |
| 18 | USBSS0_TX_P | 168 | | |

**Notes:**

1. **USB 3.2 module pin names are using the Orin NX and Orin Nano function names.**
2. **The module pin names not directly connected to the USB connector pins but are routed to the input of the USB hub.**
3. **In the Type/Dir column, Output is to USB connectors. Input is from USB connectors. Bidir is for bidirectional signals.**

| **Legend** | Ground | Power | Reserved |
|---|---|---|---|

# 2.2 Gigabit Ethernet

The carrier board implements an RJ45 connector (J15) along with the necessary magnetics device.

### Table 2-4. Ethernet RJ45 Connector Pin Description – J15

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| 1 | GPE_MDI0_P | 186 | Gigabit Ethernet MDI 0+ | Bidir |
| 2 | GPE_MDI0_N | 184 | Gigabit Ethernet MDI 0– | Bidir |
| 3 | GPE_MDI1_P | 192 | Gigabit Ethernet MDI 1+ | Bidir |
| 4 | – | – | MCT | – |
| 5 | – | – | MCT | – |
| 6 | GPE_MDI1_N | 190 | Gigabit Ethernet MDI 1– | Bidir |
| 7 | GPE_MDI2_P | 198 | Gigabit Ethernet MDI 2+ | Bidir |
| 8 | GPE_MDI2_N | 196 | Gigabit Ethernet MDI 2– | Bidir |
| 9 | GPE_MDI3_P | 204 | Gigabit Ethernet MDI 3+ | Bidir |

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| 10 | GPE_MDI3_N | 202 | Gigabit Ethernet MDI 3– | Bidir |
| 11 | – | – | Power-Over-Ethernet | Power |
| 12 | | | | |
| 13 | | | | |
| 14 | | | | |
| 15 | – | – | Green LED Anode | Input |
| 16 | GBE_LED_LINK | 188 | Green LED Cathode. On for 1,000 Mbps link. Off for 10/100Mbps. | Output |
| 17 | – | – | Yellow LED Anode | Input |
| 18 | GBE_LED_ACT | 194 | Yellow LED Cathode.  On indicates activity. | Output |
| 19 | – | – | Shield Ground | Ground |
| 20 | | | | |

**Note: In the Type/Dir column, Output is to RJ45 connector. Input is from RJ45 connector. Bidir is for bidirectional signals.**

| **Legend** | Ground | Power | Reserved |
|---|---|---|---|

# 2.3 DisplayPort

A DisplayPort (DP) connector (J8) is provided. Dual mode is supported.

### Table 2-5. DisplayPort Connector Pin Description – J8

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| 1 | DP0_TXD0_P | 41 | DP Lane 0+ | Output |
| 2 | – | – | Ground | Ground |
| 3 | DP0_TXD0_N | 39 | DP Lane 0– | Output |
| 4 | DP0_TXD1_P | 47 | DP Lane 1+ | Output |
| 5 | – | – | Ground | Ground |
| 6 | DP0_TXD1_N | 45 | DP Lane 1– | Output |
| 7 | DP0_TXD2_P | 53 | DP Lane 2+ | Output |
| 8 | – | – | Ground | Ground |
| 9 | DP0_TXD2_N | 51 | DP Lane 2– | Output |
| 10 | DP0_TXD3_P | 59 | DP Lane 3+ | Output |
| 11 | – | – | Ground | Ground |
| 12 | DP0_TXD3_N | 57 | DP Lane 3– | Output |

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| 13 | – | – | MODE to support Dual-role mode. Connects from DP connector to PI3AUX221ZTAEX device to select between DP or HDMI mode. | Input |
| 14 | – | – | CEC_DP: Not used – pulled to GND through 1 Mohm resistor | Unused |
| 15 | DP0_AUX_N | 90 | DisplayPort Auxiliary Channel 0- | Bidir |
| 16 | – | – | Ground | Ground |
| 17 | DP0_AUX_P | 92 | DisplayPort Auxiliary Channel 0+ | Bidir |
| 18 | DP0_HPD | 88 | HDMI™ Hot Plug Detect | Input |
| 19 | – | – | Power Return (Ground) | Ground |
| 20 | – | – | +3.3 V | Power |

**Note: In the Type/Dir column, Output is to DP connector. Input is from DP connector. Bidir is for bidirectional signals.**

| Legend | Ground | Power | Reserved |
|---|---|---|---|

# 2.4 M.2 Key E Expansion Slot

The Jetson Orin Nano carrier board includes a M.2, Key E Slot Mini-PCIe Expansion slot (J10). This includes interface options for WiFi/BT including PCIe (x1), USB 2.0, UART, I2S, and I2C optional.

**Notes:**

- The Jetson Orin Nano Developer Kit carrier board will only support single sided M.2 Key E modules.
- Stuffing resistors for connecting I2C2 to pins 58 and 60 of the M.2 Key E connector are not installed by default. If I2C is required, 0Ω resistors can be installed at locations R106 and R107. Care should be taken as some M.2 cards can cause conflicts with other devices connected to the I2C interface.

## Table 2-6. M.2 Key E Expansion Slot Pin Description – J10

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir | Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|---|---|---|---|---|
| 1 | – | | Ground | Ground | | – | – | No Pin | – |
| 3 | USB2_D_P | 123 | USB 2.0 Data | Bidir | 2 | – | – | Main 3.3V Supply | Power |
| 5 | USB2_D_N | 121 | | | 4 | | | | |
| 7 | – | | Ground | Ground | 6 | – | – | Unused | Unused |
| 9 | – | – | Unused | Unused | 8 | I2S1_CLK | 226 | I2S #1 Clock | Bidir, 1.8V |

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir | Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|---|---|---|---|---|
| 11 | | | | | 10 | I2S1_FS | 224 | I2S #1 Left/Right Clock | Bidir, 1.8V |
| 13 | | | | | 12 | I2S1_DIN | 222 | I2S #1 Data In | Input, 1.8V |
| 15 | | | | | 14 | I2S1_DOUT | 220 | I2S #1 Data Out | Bidir, 1.8V |
| 17 | | | | | 16 | – | – | Unused | Unused |
| 19 | | | | | 18 | – | – | Ground | Ground |
| 21 | | | | | 20 | GPIO02 | 124 | Bluetooth #2 Wake AP | Input, 3.3V |
| 23 | | | | | 22 | UART0_RXD | 101 | UART #0 Receive | Input, 1.8V |
| 25 | | | | | 24 | | | | |
| 27 | – | – | Key | Unused | 26 | – | – | Key | Unused |
| 29 | | | | | 28 | | | | |
| 31 | | | | | 30 | | | | |
| 33 | – | – | Ground | Ground | 32 | UART0_TXD | 99 | UART #0 Transmit | Output, 1.8V |
| 35 | PEX1_TX0_P | 174 | PCIe #1 Transmit Lane 0 | Output | 34 | UART0_CTS* | 105 | UART #0 Clear to Send | Input, 1.8V |
| 37 | PEX1_TX0_N | 172 | | | 36 | UART0_RTS* | 103 | UART #0 Request to Send | Output, 1.8V |
| 39 | – | – | Ground | Ground | 38 | | | | |
| 41 | PEX1_RX0_P | 169 | PCIe #1 Receive Lane 0 | Input | 40 | | | | |
| 43 | PEX1_RX0_N | 167 | | | 42 | | | | |
| 45 | – | – | Ground | Ground | 44 | – | – | Unused | Unused |
| 47 | PEX1_CLK_P | 175 | PCIe #1 Reference clock | Output | 46 | | | | |
| 49 | PEX1_CLK_N | 173 | | | 48 | | | | |
| 51 | – | – | Ground | Ground | 50 | CLK_32K_OUT | 210 | Suspend Clock (32 KHz) | Output, 3.3V |
| 53 | PEX1_CLKREQ* | 182 | PCIe #1 Clock Request | Bidir, 3.3V | 52 | PEX0_RST* | 183 | PCIe #0 Reset | Output, 3.3V |
| 55 | PEX_WAKE* | 179 | PCIe Wake | Input, 3.3V | 54 | GPIO3 | 126 | BT Enable | Output, 3.3V |
| 57 | – | – | Ground | Ground | 56 | GPIO5 | 128 | Wi-Fi Disable | Output, 3.3V |
| 59 | – | – | Unused | Unused | 58 | I2C2_SDA | 234 | General I2C #2 (optional) | Bidir/OD, 1.8V |
| 61 | | | | | 60 | I2C2_SCL | 232 | | |
| 63 | – | – | Ground | Ground | 62 | GPIO10 | 212 | M.2, Key E Connector Alert | Input, 1.8V |

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir | Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|---|---|---|---|---|
| 65 | – | – | Unused | Unused | 64 | – | – | Unused | Unused |
| 67 | | | | | 66 | | | | |
| 69 | – | – | Ground | Ground | 68 | | | | |
| 71 | – | – | Unused | Unused | 70 | | | | |
| 73 | | | | | 72 | – | – | Main 3.3V Supply | Power |
| 75 | – | – | Ground | Ground | 74 | | | | |

Note: In the Type/Dir column, Output is to M.2 module. Input is from M.2 module. Bidir is for bidirectional signals.

| Legend | Ground | Power | Reserved |
|---|---|---|---|

# 2.5 M.2 Key M Expansion Slot

The carrier board includes two M.2, Key M Slots for NVMe storage (J11 and J24). The M.2, Key M slot J11 supports PCIe (x4), Gen4. The M.2, Key M slot J24 supports PCIe (x2), Gen4.

**Notes:**

- Jetson Orin Nano modules supports only up to Gen3 PCIe, and Jetson Orin NX modules support up to Gen4 PCIe
- The Jetson Orin Nano Developer Kit carrier board will only support single sided M.2 Key M modules.

## Table 2-7. M.2 Key M Expansion Slot Pin Description – J11 (x4 PCIe)

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default | Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default |
|---|---|---|---|---|---|---|---|---|---|
| 1 | – | – | Ground | Ground | 2 | – | – | Main 3.3V Supply | Power |
| 3 | | | | | 4 | | | | |
| 5 | PCIE0_RX3_N | 155 | PCIe IF #0 Lane 3 Receive | Input | 6 | – | – | Unused | Unused |
| 7 | PCIE0_RX3_P | 157 | | | 8 | | | | |
| 9 | – | – | Ground | Ground | 10 | | | | |
| 11 | PCIE0_TX3_N | 154 | PCIe IF #0 Lane 3 Transmit | Output | 12 | – | – | Main 3.3V Supply | Power |
| 13 | PCIE0_TX3_P | 156 | | | 14 | | | | |
| 15 | – | – | Ground | Ground | 16 | | | | |
| 17 | PCIE0_RX2_N | 149 | PCIe IF #0 Lane 2 Receive | Input | 18 | | | | |
| 19 | PCIE0_RX2_P | 151 | | | 20 | – | – | Unused | Unused |

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default | Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default |
|---|---|---|---|---|---|---|---|---|---|
| 21 | – | – | Ground | Ground | 22 | | | | |
| 23 | PCIE0_TX2_N | 148 | PCIe IF #0 Lane 2 Transmit | Output | 24 | | | | |
| 25 | PCIE0_TX2_P | 150 | | | 26 | | | | |
| 27 | – | – | Ground | Ground | 28 | | | | |
| 29 | PCIE0_RX1_N | 137 | PCIe IF #0 Lane 1 Receive | Input | 30 | | | | |
| 31 | PCIE0_RX1_P | 139 | | | 32 | | | | |
| 33 | – | – | Ground | Ground | 34 | | | | |
| 35 | PCIE0_TX1_N | 140 | PCIe IF #0 Lane 1 Transmit | Output | 36 | | | | |
| 37 | PCIE0_TX1_P | 142 | | | 38 | | | | |
| 39 | – | – | Ground | Ground | 40 | – | – | Unused | Unused |
| 41 | PCIE0_RX0_N | 131 | PCIe IF #0 Lane 0 Receive | Input | 42 | | | | |
| 43 | PCIE0_RX0_P | 133 | | | 44 | GPIO10 | 212 | M.2 Key M Alert | Input, 1.8V |
| 45 | – | – | Ground | Ground | 46 | – | – | Unused | Unused |
| 47 | PCIE0_TX0_N | 134 | PCIe IF #0 Lane 0 Transmit | Output | 48 | | | | |
| 49 | PCIE0_TX0_P | 136 | | | 50 | PEX0_RST* | 181 | PCIe IF #0 Reset | Output, 3.3V |
| 51 | – | – | Ground | Ground | 52 | PEX0_ CLKREQ* | 180 | PCIe IF #0 Clock Request | Input, 3.3V |
| 53 | PCIE0_CLK_N | 160 | PCIe IF #0 Reference Clock | Output | 54 | PEX_WAKE* | 179 | PCIe Wake (Level Shifted from 3.3V to 1.8V) | Input, 3.3V |
| 55 | PCIE0_CLK_P | 162 | | | 56 | – | – | Unused | Unused |
| 57 | – | – | Ground | Ground | 58 | | | | |
| 59 | – | – | Unused (Key) | Unused | 60 | – | – | Unused (Key) | Unused |
| 61 | | | | | 62 | | | | |
| 63 | | | | | 64 | | | | |
| 65 | | | | | 66 | | | | |
| 67 | – | – | Unused | Unused | 68 | – | – | 32 KHz Suspend Clock | Output, 3.3V |
| 69 | | | | | 70 | – | – | Main 3.3V Supply | Power |
| 71 | – | – | Ground | Ground | 72 | | | | |
| 73 | | | | | 74 | | | | |
| 75 | | | | | | | – | No Pin | – |

**Note: In the Type/Dir column, Output is to M.2 module. Input is from M.2 Module. Bidir is for bidirectional signals.**

| **Legend** | Ground | Power | Reserved |
|---|---|---|---|

## Table 2-8. M.2 Key M Expansion Slot Pin Description – J24 (x2 PCIe)

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default | Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default |
|---|---|---|---|---|---|---|---|---|---|
| 1 | – | – | Ground | Ground | 2 | – | – | Main 3.3V Supply | Power |
| 3 | | | | | 4 | | | | |
| 5 | – | – | Unused | Unused | 6 | – | – | Unused | Unused |
| 7 | | | | | 8 | | | | |
| 9 | – | – | Ground | Ground | 10 | | | | |
| 11 | – | – | Unused | Unused | 12 | – | – | Main 3.3V Supply | Power |
| 13 | | | | | 14 | | | | |
| 15 | – | – | Ground | Ground | 16 | | | | |
| 17 | – | – | Unused | Unused | 18 | – | – | Unused | Unused |
| 19 | | | | | 20 | | | | |
| 21 | – | – | Ground | Ground | 22 | | | | |
| 23 | – | – | Unused | Unused | 24 | | | | |
| 25 | | | | | 26 | | | | |
| 27 | – | – | Ground | Ground | 28 | | | | |
| 29 | PCIE2_RX1_N | 58 | PCIe IF #2 Lane 1 Receive | Input | 30 | | | | |
| 31 | PCIE2_RX1_P | 60 | | | 32 | | | | |
| 33 | – | – | Ground | Ground | 34 | | | | |
| 35 | PCIE2_TX1_N | 64 | PCIe IF #2 Lane 1 Transmit | Output | 36 | | | | |
| 37 | PCIE2_TX1_P | 66 | | | 38 | | | | |
| 39 | – | – | Ground | Ground | 40 | | | | |
| 41 | PCIE2_RX0_N | 40 | PCIe IF #2 Lane 0 Receive | Input | 42 | | | | |
| 43 | PCIE2_RX0_P | 42 | | | 44 | GPIO10 | 212 | M.2 Key M Alert | Input, 1.8V |
| 45 | – | – | Ground | Ground | 46 | – | – | Unused | Unused |
| 47 | PCIE2_TX0_N | 46 | PCIe IF #2 Lane 0 Transmit | Output | 48 | | | | |
| 49 | PCIE2_TX0_P | 48 | | | 50 | PEX2_RST* | 219 | PCIe IF #0 Reset | Output, 3.3V |
| 51 | – | – | Ground | Ground | 52 | PEX2_CLKREQ* | 221 | PCIe IF #0 Clock Request | Input, 3.3V |
| 53 | PCIE2_CLK_N | 52 | PCIe IF #2 Reference Clock | Output | 54 | PEX_WAKE* | 179 | PCIe Wake (Level Shifted from 3.3V to 1.8V) | Input, 3.3V |
| 55 | PCIE2_CLK_P | 54 | | | 56 | – | – | Unused | Unused |
| 57 | – | – | Ground | Ground | 58 | | | | |
| 59 | – | – | Unused (Key) | Unused | 60 | – | – | Unused (Key) | Unused |

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default | Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default |
|---|---|---|---|---|---|---|---|---|---|
| 61 | | | | | 62 | | | | |
| 63 | | | | | 64 | | | | |
| 65 | | | | | 66 | | | | |
| 67 | – | – | Unused | Unused | 68 | – | – | 32 KHz Suspend Clock | Output, 3.3V |
| 69 | | | | | 70 | – | – | Main 3.3V Supply | Power |
| 71 | | | | | 72 | | | | |
| 73 | – | – | Ground | Ground | 74 | | | | |
| 75 | | | | | | | – | –No Pin | – |

Notes:

1. PCIe 2 module pin names are using the Orin NX and Orin Nano function names.
2. In the Type/Dir column, Output is to M.2 module. Input is from M.2 Module. Bidir is for bidirectional signals.

| Legend | Ground | Power | Reserved |
|---|---|---|---|

# Chapter 3. Carrier Board Custom Expansion IF Connections

The Jetson Orin Nano carrier board supports several expansion headers and connectors that have custom pinouts. The following lists the headers and connectors that have custom pinouts.

- Jetson Orin Nano Module Connector, 260-pin, SO-DIMM, 1.27 mm pitch
- Camera Connectors (x2), 22 position, Flex Connector, 0.5 mm pitch
- 40-Pin Expansion Header, 2x20, 2.54 mm pitch
- Button Header, 2x4, 2.54 mm pitch
- Optional CAN Bus header
- Fan Connector, 4-pin, 1.25 mm pitch
- Optional real-time-clock (RTC) back-up connector
- DC Power Jack
- Optional Power-over Ethernet (PoE) header, 1x4, 2.54 mm pitch
- Optional PoE backpower header, 1x, 2.54 mm pitch

## 3.1 Jetson Orin Nano Module Connector

The carrier board interfaces to the Jetson Orin Nano module using a 260-pin SODIMM connector (J2). The carrier board has a TE Connectivity 2309413-1 connector. This interfaces with the Jetson Orin Nano edge fingers. The connector pinout can be found in the *Jetson Orin Nano Product Design Guide*.

## 3.2 Camera Connector

The Jetson Orin Nano carrier board includes two 22-position flex (0.5 mm pitch) camera connectors (J20 and J219). The connector used on the carrier board is a Molex Japan Part #54548-2272.

The connectors support the following.

- J20: CSI 1 x2 lane
- J21: CSI 1 x2 lane or 1 x4 lane
- Both J20/J21: C
  - CAM_I2C, clock, and control GPIOs for the camera
  - 3.3V Supply

## Table 3-1. Camera #0 Connector Pin Description – J20

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| 1 | – | – | +3.3V | Power |
| 2 | CAM_I2C_SDA | 215 | Camera I2C. 2.2 kΩ pull-ups on module. The module CAM_I2C pins connect to an I2C mux. The camera connector #0 (J20) receives the I2C from the mux (first output). The I2C signals on the camera side of the mux have 1 kΩ pull-ups. | Bidir, 3.3V |
| 3 | CAM_I2C_SCL | 213 | | Output, 3.3V |
| 4 | – | – | Ground | Ground |
| 5 | CAM0_MCLK | 116 | Camera #0 Primary Clock | Output, 1.8V |
| 6 | CAM0_PWDN | 114 | Camera #0 Power-down | Output, 1.8V |
| 7 | – | – | Ground | Ground |
| 8 | CSI0_D1_P | 18 | CSI 0 Data 1 | Input |
| 9 | CSI0_D1_N | 16 | | |
| 10 | – | – | Ground | Ground |
| 11 | CSI0_D0_P | 6 | CSI 0 Data 0 | Input |
| 12 | CSI0_D0_N | 4 | | |
| 13 | – | – | Ground | Ground |
| 14 | CSI1_CLK_P | 11 | CSI 0 Clock | Input |
| 15 | CSI1_CLK_N | 9 | | |
| 16 | – | – | Ground | Ground |
| 17 | CSI1_D1_P | 17 | CSI 1 Data 1 | Input |
| 18 | CSI1_D1_N | 15 | | |
| 19 | – | – | Ground | Ground |
| 20 | CSI1_D0_P | 5 | CSI 1 Data 0 | Input |
| 21 | CSI1_D0_N | 3 | | |
| 22 | – | – | Ground | Ground |

**Note: In the Type/Dir column, Output is to camera module. Input is from camera module. Bidir is for bidirectional signals.**

| **Legend** | Ground | Power | Reserved |
|---|---|---|---|

## Table 3-2. Camera #1 Connector Pin Description – J21

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| 1 | – | – | +3.3V | Power |
| 2 | CAM_I2C_SDA | 215 | Camera I2C. 2.2 kΩ pull-ups on module. The module CAM_I2C pins connect to an I2C mux. The camera connector #1 (J21) receives the I2C from the mux (second output). The I2C signals on the camera side of the mux have 1 kΩ pull-ups. | Bidir, 3.3V |
| 3 | CAM_I2C_SCL | 213 | | Output, 3.3V |
| 4 | – | – | Ground | Ground |
| 5 | CAM1_MCLK | 122 | Camera #0 Primary Clock | Output, 1.8V |
| 6 | CAM1_PWDN | 120 | Camera #0 Power-down | Output, 1.8V |
| 7 | – | – | Ground | Ground |
| 8 | CSI3_D1_P | 35 | CSI 3 Data 1 | Input |
| 9 | CSI3_D1_N | 33 | | |
| 10 | – | – | Ground | Ground |
| 11 | CSI3_D0_P | 23 | CSI 3 Data 0 | Input |
| 12 | CSI3_D0_N | 21 | | |
| 13 | – | – | Ground | Ground |
| 14 | CSI2_CLK_P | 30 | CSI 2 Clock | Input |
| 15 | CSI2_CLK_N | 28 | | |
| 16 | – | – | Ground | Ground |
| 17 | CSI2_D1_P | 36 | CSI 2 Data 1 | Input |
| 18 | CSI2_D1_N | 34 | | |
| 19 | – | – | Ground | Ground |
| 20 | CSI2_D0_P | 24 | CSI 2 Data 0 | Input |
| 21 | CSI2_D0_N | 22 | | |
| 22 | – | – | Ground | Ground |

Note: In the Type/Dir column, Output is to camera module. Input is from camera module. Bidir is for bidirectional signals.

| Legend | Ground | Power | Reserved |
|---|---|---|---|

# 3.3 40-Pin Expansion Header

The Jetson Orin Nano carrier board includes a 40-pin (2x20, 2.54 mm pitch) expansion header (J12). The connector used on the carrier board is an Astron Technology (Part # 27-0169H-220-1G-H).

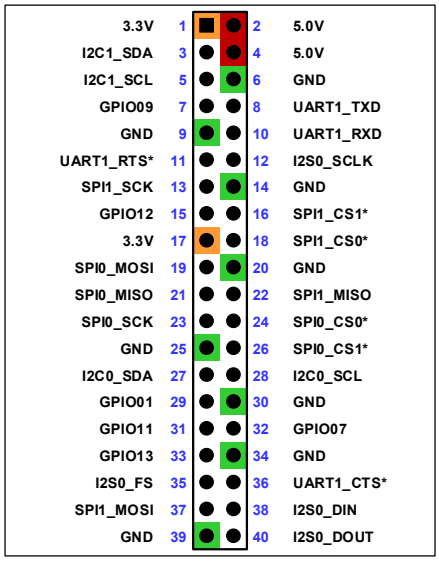
The expansion connector includes various audio and control interfaces including:

- I2S
- Audio (AU) clock
- I2C (x2)
- SPI (x2)
- UART
- GPIOs (x3 – See notes)

**Notes:**

- All the signals on the expansion header use 3.3V levels.
- All the interface signal pins (I2S, I2C, SPI, UART, and AU clock) can also be configured as GPIOs.
- Any pull-up or pull-down resistors on the signals (except I2C) must be weak (limited to >50 kΩ).

Figure 3-1. Expansion Header Connections



## Table 3-3. Expansion Header Pin Description – J12

| Header Pin # | Module Pin Name | Module Pin # | SoC Pin Name | Default Usage / Description | Alternate Functionality | Type/ Dir | Pin Drive or Power Pin Max Current | SoC GPIO Port # | Power-on Default | PU/PD on Module | Notes |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | – | – | – | Main 3.3V Supply | – | Power (input) | 1A | – | – | – | 1 |
| 2 | – | – | – | Main 5.0V Supply | – | Power (input/output) | 1A | – | – | – | 1 |
| 3 | I2C1_SDA | 191 | GP16_I2C8_DAT | I2C #1 Data | – | Bidir OD | ±2 mA | PDD.02 | z | 2.2KΩ PU | 2 |
| 4 | – | – | – | Main 5.0V Supply | – | Power | 1A | – | – | – | – |
| 5 | I2C1_SCL | 189 | GP15_I2C8_CLK | I2C #1 Clock | – | Bidir OD | ±2 mA | PDD.01 | z | 2.2KΩ PU | 2 |
| 6 | – | – | – | Ground | – | Ground | – | – | – | – | – |
| 7 | GPIO09 | 211 | GP167 | GPIO | Audio Primary Clock | Bidir/Output | ±20uA | PAC.06 | pd | | 3 |
| 8 | UART1_TXD | 203 | GP70_UART1_TXD_BOOT2_STRAP | UART #1 Transmit | GPIO | Output/Bidir | ±20uA | PR.02 | pd | | 3 |
| 9 | – | – | – | Ground | – | Ground | – | – | – | – | – |
| 10 | UART1_RXD | 205 | GP71_UART1_RXD | UART #1 Receive | GPIO | Input/Bidir | ±20uA | PR.03 | pd | | 3 |
| 11 | UART1_RTS* | 207 | GP72_UART1_RTS_N | GPIO | UART #2 Request to Send | Bidir/Output | ±20uA | PR.04 | pd | | 3 |
| 12 | I2S0_SCLK | 199 | GP122 | GPIO | Audio I2S #0 Clock | Bidir | ±20uA | PH.07 | pd | | 3 |
| 13 | SPI1_SCK | 106 | GP36_SPI3_CLK | GPIO | SPI #1 Shift Clock | Bidir/Output | ±20uA | PY.00 | pd | | 3 |
| 14 | – | – | – | Ground | – | Ground | – | – | – | – | – |
| 15 | GPIO12 | 218 | GP88_PWM1 | GPIO | – | Bidir | ±20uA | PN.01 | z | | 3 |
| 16 | SPI1_CS1* | 112 | GP40_SPI3_CS1_N | GPIO | SPI #1 Chip Select #1 | Bidir/Output | ±20uA | PY.04 | z | | 3 |
| 17 | – | – | – | Main 3.3V Supply | – | Power | 1A | – | – | – | 1 |
| 18 | SPI1_CS0* | 110 | GP39_SPI3_CS0_N | GPIO | SPI #0 Chip Select #0 | Bidir/Output | ±20uA | PZ.06 | z | | 3 |
| 19 | SPI0_MOSI | 89 | GP49_SPI1_MOSI | GPIO | SPI #0 Primary Out/Secondary In | Bidir/Output | ±20uA | PZ.05 | pd | | 3 |
| 20 | – | – | – | Ground | – | Ground | – | – | – | – | – |
| 21 | SPI0_MISO | 93 | GP48_SPI1_MISO | GPIO | SPI #0 Primary In/Secondary Out | Bidir/Input | ±20uA | PZ.04 | pd | | 3 |
| 22 | SPI1_MISO | 108 | GP37_SPI3_MISO | GPIO | SPI #1 Primary In/Secondary Out | Bidir/Input | ±20uA | PY.01 | pd | | 3 |
| 23 | SPI0_SCK | 91 | GP47_SPI1_CLK | GPIO | SPI #0 Shift Clock | Bidir/Output | ±20uA | PZ.03 | pd | | 3 |
| 24 | SPI0_CS0* | 95 | GP50_SPI1_CS0_N | GPIO | SPI #0 Chip Select #0 | Bidir/Output | ±20uA | PZ.06 | z | | 3 |

| Header Pin # | Module Pin Name | Module Pin # | SoC Pin Name | Default Usage / Description | Alternate Functionality | Type/ Dir | Pin Drive or Power Pin Max Current | SoC GPIO Port # | Power-on Default | PU/PD on Module | Notes |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 25 | – | – | – | Ground | – | Ground | – | – | – | – | – |
| 26 | SPI0_CS1* | 97 | GP51_SPI1_CS1_N | GPIO | SPI #0 Chip Select #1 | Bidir/Output | ±20uA | PZ.07 | pu | | 3 |
| 27 | I2C0_SDA | 187 | GP14_I2C2_DAT | I2C #0 Data | GPIO | Bidir OD/Bidir | ±2 mA | PDD.00 | z | 1.5KΩ PU | 2 |
| 28 | I2C0_SCL | 185 | GP13_I2C2_CLK | I2C #0 Clock | GPIO | Bidir OD/Bidir | ±2 mA | PCC.07 | z | 1.5KΩ PU | 2 |
| 29 | GPIO01 | 118 | GP65 | GPIO | General Purpose Clock #0 | Bidir/Output | ±20uA | PQ.05 | pd | | 3 |
| 30 | – | – | – | Ground | – | Ground | – | – | – | – | – |
| 31 | GPIO11 | 216 | GP66 | GPIO | General Purpose Clock #1 | Bidir/Output | ±20uA | PQ.06 | pd | | 3 |
| 32 | GPIO07 | 206 | GP113_PWM7 | GPIO | PWM | Bidir/Output | ±20uA | PG.06 | z | | 3 |
| 33 | GPIO13 | 228 | GP115 | GPIO | PWM | Bidir/Output | ±20uA | PH.00 | z | | 3 |
| 34 | – | – | – | Ground | – | Ground | – | – | – | – | – |
| 35 | I2S0_FS | 197 | GP125 | GPIO | Audio I2S #0 Field Select | Bidir | ±20uA | PI.02 | pd | | 3 |
| 36 | UART1_CTS* | 209 | GP73_UART1_CTS_N | GPIO | UART #1 Clear to Send | Bidir/Input | ±20uA | PR.05 | pd | | 3 |
| 37 | SPI1_MOSI | 104 | GP38_SPI3_MOSI | GPIO | SPI #1 Primary Out/Secondary In | Bidir/Output | ±20uA | PY.02 | pd | | 3 |
| 38 | I2S0_DIN | 195 | GP124 | GPIO | Audio I2S #0 Data in | Bidir/Input | ±20uA | PI.01 | pd | | 3 |
| 39 | – | – | – | Ground | – | Ground | – | – | – | – | – |
| 40 | I2S0_DOUT | 193 | GP123 | GPIO | Audio I2S #0 Data Out | Bidir/Output | ±20uA | PI.00 | pd | | 3 |

Notes:

1. This is current capability per power pin.
2. These pins are connected to the SoC directly. They are open-drain (either pulled up or driven low by the SoC when configured as outputs). The max drive that meets the data sheet VOL is ±2 mA.
3. These pins connect to TI TXB0108 level translators. Due to the design of these devices, the output drivers are very weak, so they can be overdriven by another connected device output for bidirectional support.
4. In the Type/Dir column, output is to expansion header. Input is from expansion header. Bidir is for bidirectional signals. Where two directions are shown, the first is for the primary function (mostly GPIOs) and the second is for the alternate function.
5. Where the signal direction is input or output in this table (Table 3-3), this matches the typical special function usage (for example, SPI, I2S, and so on). The direction is bidirectional if these are configured as GPIOs.
6. All signals on the 40-pin header are 3.3V levels.

| Legend | Ground | Power | Reserved |
|---|---|---|---|

# 3.4 Button Header

The Jetson Orin Nano carrier board brings several system signals (power, reset, and force recovery), UART, and Sleep/Wake LED-related signals to a pair of standard 0.254 mm pitch header. The button header is J14.

**Table 3-4. Button Header Description – J14**

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default |
|---|---|---|---|---|
| 1 | – | – | PC_LED-: Connects to LED Cathode to indicate System Sleep/Wake (Off when system in sleep mode) | Input, 5V |
| 2 | – | – | PC_LED+: Connects to LED Anode (see Pin 1) | Output |
| 3 | UART2_RXD (DEBUG) | 238 | UART #2 Receive | Input, 3.3V |
| 4 | UART2_TXD (DEBUG) | 236 | UART #2 Transmit | Output, 3.3V |
| 5 | – | – | AC OK: Connect pins 5 and 6 to disable Auto-Power-On and require power button press. | Input, 3.3V |
| 6 | – | – | Auto Power-on disable: Pulled to GND. See Pin 5. | Not applicable |
| 7 | – | – | Ground | Ground |
| 8 | SYS_RESET* | 239 | Temporarily connect pins 7 and 8 to reset system | Input, 1.8V |
| 9 | – | – | Ground | Ground |
| 10 | FORCE_RECOVERY* | 214 | Connect pins 9 and 10 during power-on to put system in USB Force Recovery mode. | Input, 1.8V |
| 11 | – | – | Ground | Ground |
| 12 | SLEEP/WAKE* | 240 | Connect pins 11 and 12 to initiate power-on if Auto-Power-On disabled (Pins 5 and 6 connected). | Input, 5V |

**Note: In the Type/Dir column, Output is to button header. Input is from button header. Bidir is for bidirectional signals.**

| **Legend** | Ground | Power | Reserved |
|---|---|---|---|

# 3.5 Optional CAN Bus Header

The Jetson Orin Nano carrier board includes the footprint for a 4-pin, 2.54 mm pitch header (J17) which supports a CAN bus interface.

### Table 3-5. Optional CAN Header Pin Description – J17

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default |
|---|---|---|---|---|
| 1 | CAN_TX | 145 | CAN Bus transmit | Output, 3.3V |
| 2 | CAN_RX | 143 | CAN Bus receive | Input, 3.3V |
| 3 | – | – | Ground | Ground |
| 4 | – | – | Main 3.3V Supply | Power |

**Note: In the Type/Dir column, Output is to CAN connector. Input is from CAN connector. Bidir is for bidirectional signals.**

| Legend | Ground | Power | Reserved |
|---|---|---|---|

# 3.6 Fan Connector

The Jetson Orin Nano carrier board includes a 4-pin fan header (J13). The connector used is a Singatron Enterprise Co., Ltd., Part # 2WBA2542WVC-F-04PNLBT1N00G.

### Table 3-6. Fan Connector Pin Description – J13

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default |
|---|---|---|---|---|
| 1 | – | – | Ground | Ground |
| 2 | – | – | Main 5.0V Supply | Power |
| 3 | GPIO08 | 208 | Fan Tachometer signal | Input, 5V |
| 4 | GPIO14 | 230 | Fan Pulse Width Modulation signal | Output, 5V |

**Note: In the Type/Dir column, Output is to fan connector. Input is from fan connector. Bidir is for bidirectional signals.**

| Legend | Ground | Power | Reserved |
|---|---|---|---|

# 3.7 Optional Battery Back-up Coin Cell Holder

The Jetson carrier board has a connector where an RTC backup battery can be connected. This connector (J3) is commonly used on laptop designs. The connector used is a Wieson Technologies AC2651-0011-003-HH, 2-pin, 1.25 mm pitch connector.

**Table 3-7. RTC Backup Batter Connector Pin Description – J3**

| Pin # | Associated Module Pin Name | Module Pin # | Usage/Description | Type/Dir |
|---|---|---|---|---|
| 1 | PM<br>IC_BBAT | 235 | RTC backup battery supply | Power |
| 2 | – | – | Ground | Ground |

| Legend | Ground | Power | Reserved |
|---|---|---|---|

# 3.8 DC Power Jack

The Jetson Orin Nano carrier board uses a DC power jack (J16) to bring in the power from the included DC power supply. The jack used on the carrier board is a Singatron Enterprise part (part #: 2DC-0005D206F). The mating barrel jack connector dimensions are:

- Barrel length: 9.5 mm
- Barrel diameter: 5.5 mm
- Pin receptacle: Accepts 2.5 mm jack pin
- The center pin is positive (+V)
- Max current supported is 3.5 A

**Figure 3-2. Jack Connector**

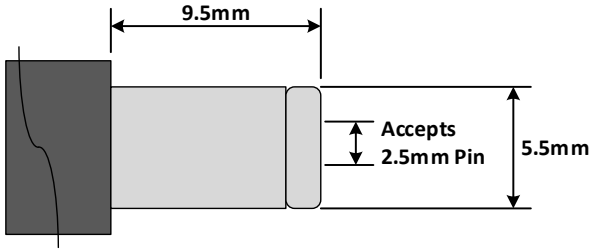


Table 3-8. DC Jack Pin Description – J16

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default |
|---|---|---|---|---|
| 1 | – | – | Main DC input supplying DC jack input (9-20 V) | Power |
| 2 | – | – | Ground | Ground |
| 3 | – | – | Ground | Ground |

| Legend | Ground | Power | Reserved |
|---|---|---|---|

# 3.9 Optional Power-Over Ethernet and Backpower Headers

The Jetson Orin Nano carrier board provides an option for an alternate main power input (besides the DC power jack). A 4-pin Power over Ethernet (PoE) header (J19 – 1x4 male, 2.54 mm pitch) brings out the VC power pins of the Ethernet connector. In addition, a 2-pin Backpower header (J18 – 1x2 male, 2.54 mm pitch) provides an alternate path for the input voltage (3 A max). To use this alternate PoE power mechanism, the design will require a power converter to take the high-voltage PoE supply (38 V-60 V) and convert it to the 5 V-20 V input the carrier board requires.

Figure 3-3. PoE Alternative Power Input

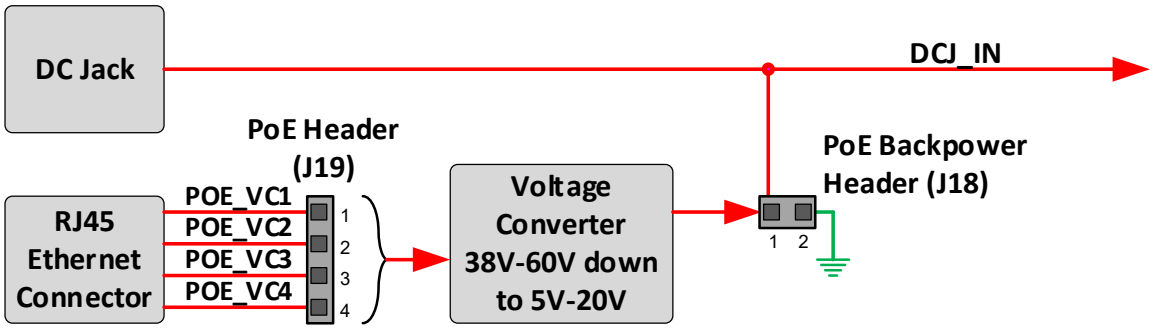



Table 3-9. PoE Header – J19

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default |
|---|---|---|---|---|
| 1 | – | – | Ethernet RG45 connector PoE VC1 power | Power |
| 2 | – | – | Ethernet RG45 connector PoE VC2 power | Power |
| 3 | – | – | Ethernet RG45 connector PoE VC3 power | Power |
| 4 | – | – | Ethernet RG45 connector PoE VC4 power | Power |

| Legend | Ground | Power | Reserved |
|---|---|---|---|

## Table 3-10. PoE Backpower Header – J18

| Pin # | Module Pin Name | Module Pin # | Usage/Description | Type/Dir Default |
|---|---|---|---|---|
| 1 | – | – | Main DC input supplying DC jack input (9 V-20 V). 3 A max. | Power |
| 2 | – | – | Ground | Ground |

**Legend**

| Ground | Power | Reserved |
|---|---|---|

# Chapter 4. Mechanical Specification

Figure 4-1 and Figure 4-2 show the mechanical dimensions for the carrier board and the developer kit.
The Developer Kit Weighs 0.175kg

Figure 4-1. Developer Kit Carrier Board Mechanical Dimensions

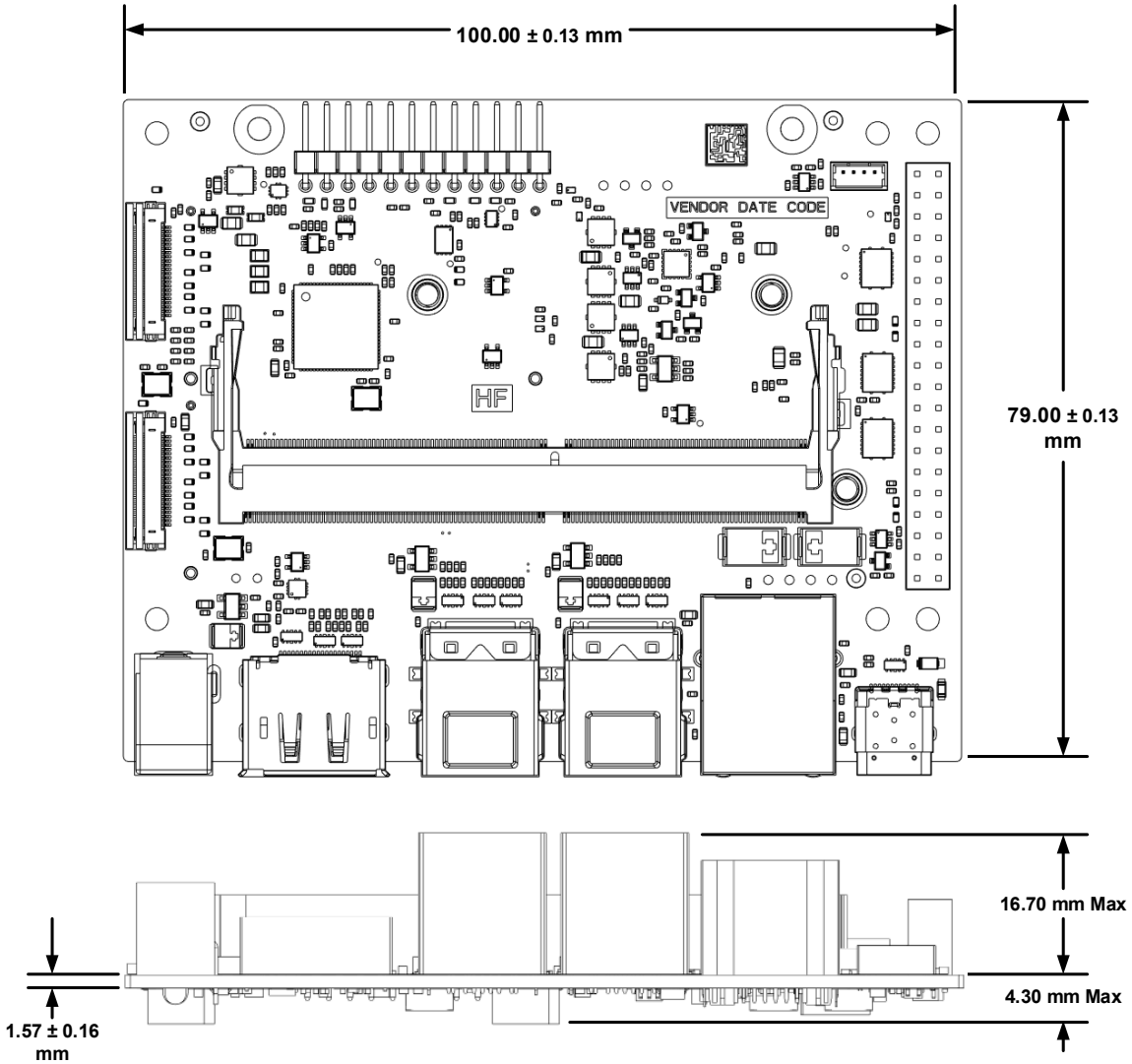
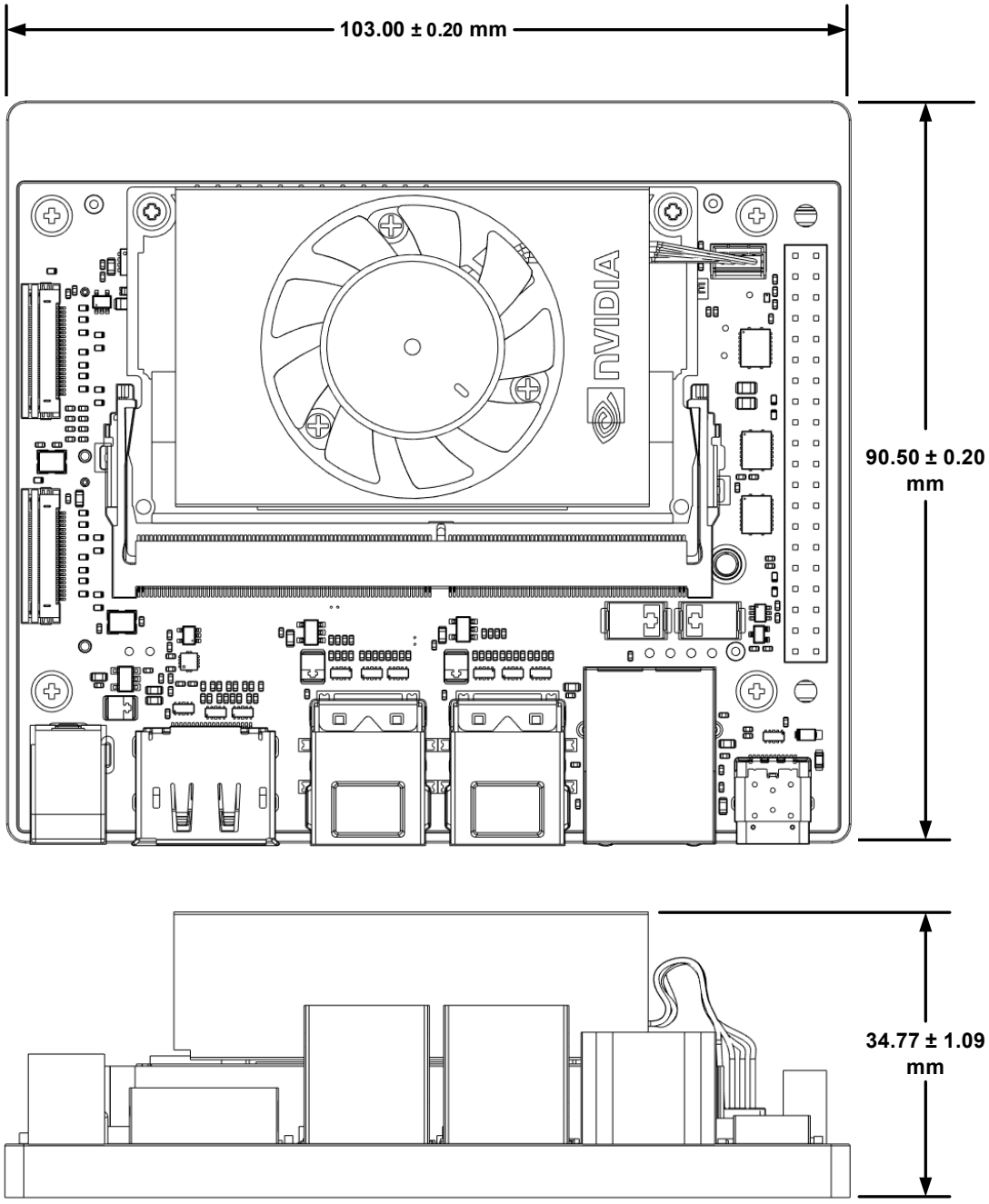
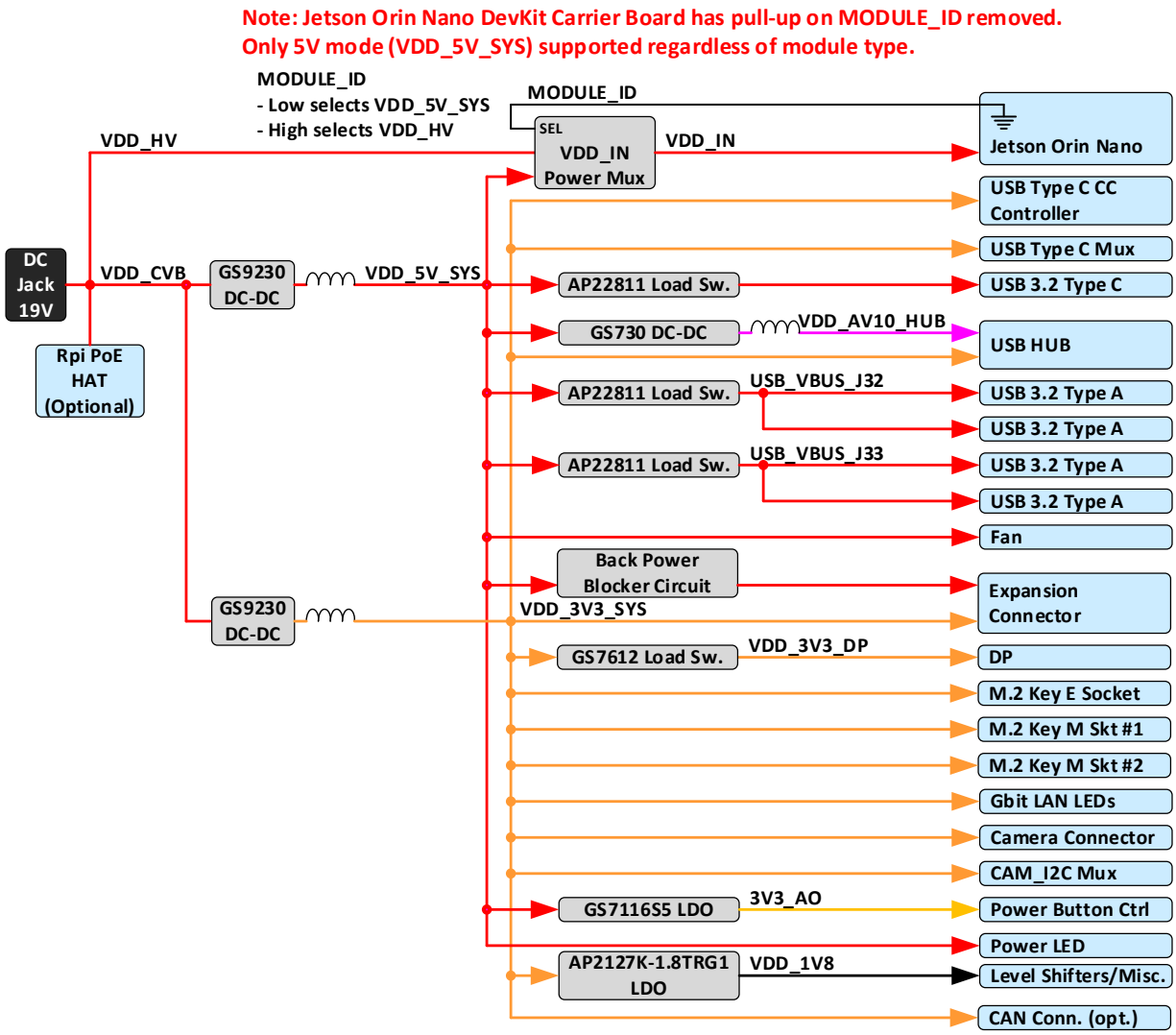


Figure 4-2. Developer Kit Mechanical Dimensions



# Chapter 5. Interface Power

Figure 5-1. Power Diagram



The following tables show the allocation of supplies to the connectors on the Jetson Orin Nano carrier board and current capabilities.

## Table 5-1. Interface Power Supply Allocation

| Power Rails | Usage | (V) | Power Supply or Gate | Source | Enable |
|---|---|---|---|---|---|
| DC_IN | Main power input from DC Adapter | 19.5 | AONR21357025 Power mux | DC Adapter | |
| VDD_5V_SYS | Main 5.0V supply | 5.0 | GS9230NVTQ-R | VDD_CVB (DC_IN after Power FET) | VDD_CVB |
| VDD_3V3_SYS | Main 3.3V supply | 3.3 | GS9230NVTQ-R | VDD_CVB | SYS_RESET_IN* |
| 3V3_AO | Button MCU | 3.3 | GS7116S5-ADJ-R LDO | VDD_5V_SYS | VDD_5V_SYS regulator power good |
| VDD_1V8 | Main 1.8V supply | 1.8 | AP2127K-1.8TRG1 | VDD_3V3_SYS | 3.3V_IO_PG |
| VDD_AV10_HUB | USB hub | 1.1 | GS7303ACTD-R | VDD_5V_SYS | VDD_3V3_SYS regulator power good |
| USBC_VBUS | 5V VBUS for USB Type C connector | 5.0 | AP22811AW5-7 | VDD_5V_SYS | ID from USB Type C CC controller |
| USB_VBUS_A | 5V VBUS for dual stacked 3.0 Type A connector | 5.0 | AP22811AW5-7 Load Switch | VDD_5V_SYS | From USB Hub |
| USB_VBUS_B | 5V VBUS for dual stacked 3.0 Type A connector | 5.0 | AP22811AW5-7 Load Switch | VDD_5V_SYS | From USB Hub |
| VDD_3V3_DP | 3.3V rail for DP connector | 5.0 | GS7612S5MC-R Load Switch | VDD_3V3_SYS | VDD_3V3_SYS regulator power good |

## Table 5-2. Interface Supply Current Capabilities

| Power Rails | Usage | (V) | Max Current (A) |
|---|---|---|---|
| DCJ_IN | Main power input from DC Adapter | 19.0 | 4.2 |
| VDD_5V_SYS | Main 5.0V supply | 5.0 | 7.8 |
| VDD_3V3_SYS | Main 3.3V supply | 3.3 | 5.4 |
| VDD_1V8 | Main 1.8V supply | 1.8 | 0.0002 |
| 3V3_AO | 3.3V Always-on supply | 3.3 | 0.200 |

## Table 5-3. Supply Current Capabilities per Connector per Supply

| Power Rails | Connector | (V) | Max Current (A) |
|---|---|---|---|
| VDD_5V_SYS | SO-DIMM (VDD_IN) | 5.0 | 5.0 |
| | 40-pin header | | 0.5 |
| | Fan connector | | 0.15 |
| VDD_3V3_DP | DP connector | | 0.5 |
| USBC_VBUS | USB Type C | | 0.5 |
| USB_VBUS_A/B | USB 3.2 type A (x4) | | 0.5 |
| VDD_3V3_SYS | 40-pin header | 3.3 | 0.1 |
| | M.2 Key E socket | | 0.8 |
| | M.2 Key M sockets (x2) | | 2.1 |
| | Camera connectors (x2) | | 0.26 |

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